HSLU
Hochschule Luzern
Departement Informatik

Project Proposal
DSPRO2

Proposal for Data Science Project

Semester: FS26

Project Title	PFAS-trace
Course / Module	DSPRO2 FS26
Team Members	Björn Pfoster, Boris Lauper
Supervisor / Coach	Ph D. Safouane El Ghazouali
Date of Submission	02.10.2026

1 Short Project Description

1.1 Problem

PFAS are released into the environment through a few key industrial processes and pose a serious health risk to the public.

1.2 Motivation

It is difficult to convey the extent of the problem on an individual level. Therefore this project aims to find an abstract correlation between the PFAS measurements and the surrounding industries.

1.3 Research Question

Can proximity to different industrial sectors explain spatial variation in PFAS concentrations and PFAS composition in French environmental measurements?

1.4 Project Objective

Build a spatial model that investigates whether the presence and type of nearby industrial facilities are associated with PFAS concentrations and PFAS profiles in environmental measurements in France.

2 Data Description

Early 2023, the Forever Pollution Project showed that nearly 23,000 sites all over Europe are contaminated by the forever chemicals PFAS. This unique collaborative cross-border and cross-field investigation by 16 European newsrooms revealed an additional 21,500 presumptive contamination sites due to current or past industrial activity. PFAS contamination spreads all over Europe.

In January 2024, the Centre national de la recherche scientifique (CNRS) compiled a large dataset based on the Forever Pollution Project's data containing all available PFAS measurements across Europe. This dataset was visualized using a map in the PFAS Data Hub (PDH) project. The creation of the dataset is documented on the PDH data processing page.

This data source can be retrieved and used for scientific purposes.

2.1 Dataset

The dataset is retrieved from the PFAS Data Hub map. It contains approximately 1'318'803 records, before filtering to France.

- **Feature types:** geolocation, concentration of the measured chemical, measurement date (and time where available), surrounding industry, and the original data source.
- **Target variable:** PFAS concentration at a given geographic location split up by chemical.
- **Explanatory variables:** presence and type of nearby industrial facilities, including known PFAS producers.

2.1.1 Preprocessing Needs

Based on the initial data assessment, some preprocessing needs are apparently needed. The dataset contains all available measurements for the continent of Europe. This project focuses on France only due to the abundant measurements compared to other countries like Spain or Germany therefore other countries and records have to be removed.

2.1.2 Quality Constraints

The data was gathered from over 100 sources and then processed by the scientists from CNRS. Not all entries are complete, most of them outside of France.
Because locations of PFAS users and PFAS producers where no measurements have yet been taken were also included in the dataset to complete the map, the dataset contains missing measurement data for these records.

3 Cloud Service Integration

Expected Production Environment
The inference Pipeline will be run on a cloud compute platform for performance and availability reason. The deployment of the production models will be done using GitHub Actions.

4 Kanban Tool

GitHub projects for progress tracking, split into pending, in progress and done.

5 Experiment Tracking Tool

Experiments will be tracked using MLFlow, an open-source platform for managing the machine learning lifecycle. The following MLFlow features are intended to be used:

- **Tracking:** Every training run records its parameters, metrics and output files, so that runs are reproducible and can be compared in the MLflow UI.
- **Model Registry:** Trained models are versioned centrally. Aliases (e.g. `champion`, `challenger`) mark which version the inference pipeline on cloud compute platform should load. Models are stored with their dependencies and input signature, so that the same artifact can be loaded for training evaluation and production inference.

6 Use of Generative AI

According to the project guidelines of the HSLU Data Science Project Module, the use of AI is allowed. The used Generative models are listed below including the intended project usage:

- **ChatGPT**: Used for brainstorming and text improvements.
- **Gemini**: Used for brainstorming and text improvements.
- **Claude**: Used for brainstorming, text improvements and code generation. Code generation is done by providing the required outcome to the model including the intended code structure. No complete hand-off of architectural or code design work will be done.